CONSTANT-pH MOLECULAR DYNAMICS OF BIOLOGICAL LIPIDS

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ABSTRACT

Negatively charged phospholipids are ubiquitous membrane components playing important roles in signalling pathways, by regulating cell apoptosis, cell replication, immune evasion in cancer cells and stress responses in plants. Importantly, lipids protonation is sensitive to different electrostatic stimuli (i.e., pH, ion concentration, membrane environment, interactions with proteins or other biomolecules) allowing the membrane to be a highly electrostatically responsive element. Unfortunately, less is known about the molecular forces that drive protonation dynamics of titratable lipids in biological membrane and how it affects membrane characteristics.

Using microsecond-long atomistic constant-pH molecular dynamics (CpHMD) simulations, we investigate titratable membrane lipids typical of eukaryotic membranes (e.g., phosphatidylserine, phosphatidic acid, phosphatidylinositols, and pyrophosphate lipids) to elucidate how their protonation dynamics are influenced by the local electrostatic environment and, in turn, how their protonation states shape membrane properties.

In our previous work involving synthetic aminolipids (1), we showed that CpHMD accurately reproduces experimentally observed pKa shifts of 3–4 units, influenced by charge repulsion, different concentrations of cholesterol and hydrogen bonding interactions with neighbouring lipids. Here, we demonstrate that CpHMD quantitatively reproduces experimental apparent pKa values for biologically relevant lipid mixtures, providing a realistic description of the pH-dependent behaviour of biological membranes. In addition to the atomistic insights into the biology of titratable lipids, our work establishes a comprehensive library of constant-pH lipid models covering the major titratable lipid species found in eukaryotic membranes, thereby enabling realistic simulations of complex biological membranes across a wide range of pH and environmental conditions.

REFERENCES

1 *Trollmann, M. F. W.; Rossetti, P.; Böckmann, R. A. Membrane Environment Sets the Functional pKa of Ionizable Lipids. Biophys. J. 2026, 0 (0). https://doi.org/10.1016/j.bpj.2026.06.030.*